Can Social Networks Predict Running Behavior? Social Aggregation and Graph Neural Networks on Strava

Giacomo Faccin GIACOMO.FACCIN905@GMAIL.COM
Università Ca' Foscari Venezia, Italy

Abstract

Strava is one of the most used sport platforms and social networks among athletes, and for this reason it was chosen to conduct a sociological analysis. More specifically, the study aims to investigate Strava as a social infrastructure for aggregation among runners, and to examine to what extent the running behavior of athletes can be predicted using information about their social network. The analysis uses the data from the study of Franken et al. (2022a). The study analyses 329 athletes from five running clubs. Combining social network analysis with Graph Neural Networks (GNNs) Hamilton et al. (2017a), it explores the structure of the running community and whether social connections improve the prediction of running behavior. The network analysis shows that kudos form cohesive and reciprocal communities, and that once the number of ties is taken into account, high-status athletes are neither more embedded nor less bridging than other athletes: the apparent difference in bridging is explained by the many low-status athletes with a single tie. The predictive analysis finds that contacts do not add information beyond the athlete's own recent history. The question of whether high-status athletes diffuse running to their contacts is only partially addressed, since the predictive models do not test the role of contact status directly. The results should therefore be read as an indication of the limits of social influence in this setting, not as evidence of its absence.

Keywords: List of keywords

1. Introduction

Digital fitness platforms have become central spaces for sport participation, and their social dimension has so far been studied mainly from a psychological perspective. Franken et al. (2022b) analyzed how recreational runners benefit from running with others to maintain a consistent training regimen over time. A study on collegiate club runners Russell et al. (2023) looked instead at the psychosocial implications of using Strava, and showed that the app shapes athletes' experience along three main themes: self-presentation, social pressure and motivation.

These works describe how athletes perceive the platform Zeng et al. (2018), but they do not observe how behavior changes inside the network of contacts. This question was addressed by Franken et al. (2022a), who studied how runners in the same virtual Strava club influenced each other. They assumed that receiving kudos on recorded activities would increase exercise rates, and that the running behavior of the clubmates to whom ego gave kudos would work as a motivational example. To test these hypotheses they used a SIENA model Snijders et al. (2010), a stochastic actor-oriented approach.

In this study we use the same data but we do not estimate the influence of kudos. We look at how kudos organize the clubs and where the high-status athletes are located in the network, since a tie between separate groups can carry a behavior across the club Granovetter (1973); Burt (2004), while ties inside dense groups can reinforce it through repeated exposure Centola (2010). The structure of the networks is described and the position of athletes and the similarity between connected ones are assessed with permutation tests McPherson et al. (2001). After that we check whether running behavior can be predicted, and whether knowing an athlete's contacts improves the prediction, comparing gradient-boosted trees Chen and Guestrin (2016) with a graph neural network Hamilton et al. (2017b). Control experiments with no edges and with randomly rewired edges are used to isolate the contribution of the contacts.

© G. Faccin.

2. Methods

Note:
In this section of study we are explaining the methodology and the procedure to produce our analysis.

2.1. Data Description

Data used in this study were obtained from the reproduction package of Franken et al. (2022a). The data are stored in the *RSiena* format and cover **five Strava running clubs, with 329 athletes, observed monthly during 2019** across 12 waves. For every athlete and month, the data report running frequency and volume, other sporting activity, and the *kudos* exchanged with the other members of the club.

The raw data were transformed into analysis-ready tables with two helper modules, which must be placed in the same folder as the notebook. The function `load_clubs` reads the export folder and returns, for each club, the athletes, the monthly kudos array (athlete × athlete × 12 waves) and the behavioral matrices (athlete × 12 waves). The function `build_frames` then converts these arrays into two long-format tables: a panel with one row per athlete and month, and an edge list with one row per kudos tie and month.

2.2. Panel Data Creation

The function `load_clubs` returns the athlete-level and kudos arrays for each club, which `build_frames` converts into one panel dataset and one edge table per club. To ensure a reliable and efficient workflow, the club-specific panel datasets and edge tables were then merged into a single panel dataset and a single edge table. The resulting panel dataset comprises 3,948 athlete-month observations, corresponding to 329 athletes observed over 12 monthly waves. The athletes are distributed across five running clubs, whose sizes are reported in Table 1.

Table 1: Number of athletes per running club

Club	Athletes
1	27
2	58
3	159
4	9
5	76
Total	329

As shown in Table 1, the clubs differ considerably in size. Clubs 1 and 4, with 27 and 9 athletes, respectively, are included in the estimation, but their results are given limited weight in the interpretation, since the small number of athletes makes the estimates for these clubs less precise.

2.3. Explanatory Data Analysis

Table 2 reports the number of athlete-month observations and the mean of the main variables for each club. Mean running frequency ranges from 1.71 (club 4) to 3.52 (club 5), and mean running volume from 1.94 to 3.43. Mean kudos given and received range from 2.00 (club 3) to 11.05 (club 2), and their distribution is right-skewed in every club. Mean Strava tenure lies between 4.5 and 6.2. Running and kudos means refer to valid observations. Running volume for club 5 is measured on a wider scale (0–9), so its values are not directly comparable with those of the other clubs.

Table 2: Descriptive statistics by club

	C1	C2	C3	C4	C5
Athletes	27	58	159	9	76
Observations	324	696	1908	108	912
Running frequency	2.32	3.03	2.27	1.71	3.52
Running volume	2.51	3.10	2.23	1.94	3.43
Kudos given/received	2.73	11.05	2.00	2.23	7.92
Strava tenure	5.5	4.5	5.7	6.2	5.7

Note. Means of running and kudos variables are computed on valid observations. Club 5 uses a wider scale for running volume (0–9). Strava tenure is measured at the athlete level and is not a measure of running experience.

Figure 1 presents two histograms with the month-to-month changes of `freq_run` and `time_run`. On the x axis the changes are clipped between −3 and +3 to highlight the change in the variables, so the last bars also contain the larger changes. Observing the figure we can deduce that in 57% of the athlete-month transitions the athletes did not change category in `freq_run` and `time_run`. Both variables are weekly categories, not measures of performance, so "no change" means that the athlete stayed in the same category, not that the activity was identical. We also need to consider that the stable differences between athletes account for about 74% of the variation in running frequency, while the rest reflects month-to-month change.

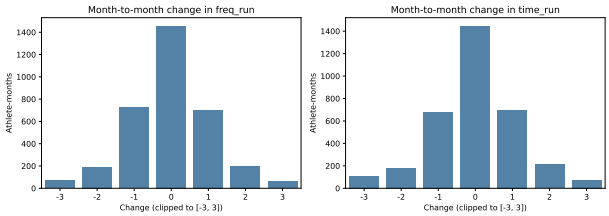


Figure 1: Month-to-month change in running frequency (left) and running volume (right) across athlete-months. Changes are clipped to $[-3, 3]$. A value of zero indicates that the athlete's category did not change between consecutive valid months.

2.4. Network Analysis

Table 3 summarizes the three graphs used in the network analysis. The first is the kudos graph, a directed and weighted graph built over the whole year (12 monthly waves). Each athlete is a node, and an arc from athlete A to athlete B means that A gave at least one kudos to B in at least one month; the weight of the arc is the number of months in which this happened, from 1 to 12. A separate graph is built for each club, and all members are included as nodes, also those without any tie. Each node carries the commitment level of the athlete, gender, tenure on Strava, an intensity index of running, and the mean number of partners to whom the athlete gave kudos, and from whom received them, per month.

The reciprocal graph is undirected and weighted, and keeps only the strong ties: two athletes are linked when they gave kudos to each other in the same month, and the weight counts how many months this happened. We use it for local bridges and embeddedness.

For the predictive part we build one snapshot graph for each club and month, from the first to the eleventh (the twelfth has no next month to predict). An edge is a kudos given in that month, and we make edges symmetric before giving them to the graph neural network, so the direction of the kudos is lost. Each snapshot contains only information available up to month t. Commitment level and status are computed over the whole year, so they appear in the descriptive analysis only.

For each club we plot the kudos graph aggregated over the year (Figure 2). In the same picture we can see the commitment level of the athletes, how much recognition they receive and how long the ties last. There is one figure per club.

Then we look at the degree distribution. Figure 2 shows it for Club 1, together with the largest weakly connected component, where the direction of the arcs is ignored. The distribution is shown in three ways: rank plot, histogram and $P(k)$. The degree is in-degree plus out-degree, so a partner who exchanges kudos with an athlete in both directions is counted twice.

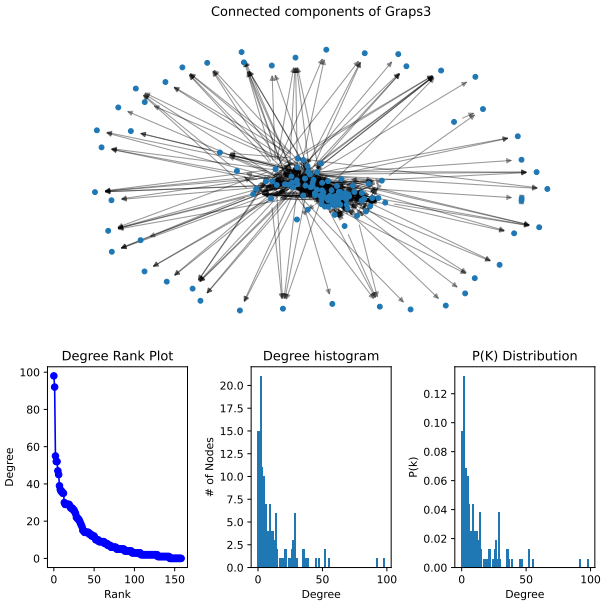
The structure of each club is then summarized with standard network measures, computed on two graphs. The kudos graph shows recognition, also when the kudos are not returned. The reciprocal graph keeps only the ties where two athletes gave kudos to each other in the same month, and these are what we call strong ties. Looking at the two together tells us how much of the visible activity becomes a real exchange, which is the first part of our question on Strava as an infrastructure for social aggregation.

2.5. Embeddedness and Strong Ties

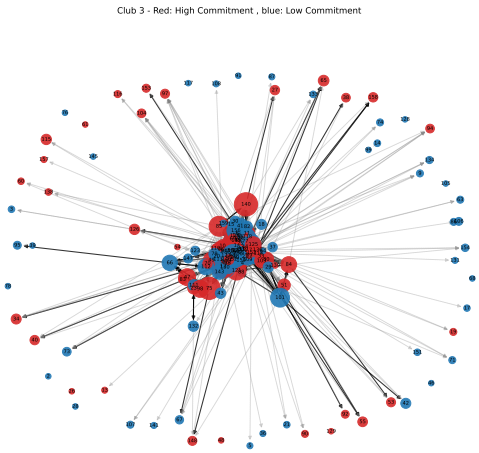
The second part of the analysis concerns high status athletes and the contacts they share activities with. For this part we use the reciprocal graph, so

Table 3: The three graphs built from the club-by-month kudos table.

Graph	Direction	Time	Tie	Used for
Kudos graph	directed, weighted	whole year	A gave at least one kudos to B in some month (weight = number of months)	recognition and status (in-degree), ego network, degree distribution, homophily matrix
Reciprocal graph	undirected, weighted	whole year	A and B exchanged kudos in the same month (weight = number of such months)	strong ties: local bridges, embeddedness
Monthly snapshots	symmetrised	month t	kudos given in that month	GNN input, information up to t



(a) Degree analysis: layout (top), rank plot, histogram and $P(k)$ (bottom).



(b) Kudos network colored by commitment level (red: high, blue: low) and sized by mean kudos received.

Figure 2: Club 3 kudos network. Solid edges indicate ties lasting at least three months, dashed gray edges shorter-lived ties. The degree distribution is strongly right-skewed, with a few hubs close to degree 100.

only strong ties are considered, and we compare the athletes along two dimensions: status, defined as in-degree in the annual kudos graph above the club median, and commitment level.

The first step was to identify the local bridges. A local bridge is a strong tie whose two ends have no common contact in the club. It connects parts of the club that are otherwise separate, and it is the typical channel through which something new reaches a group.

The second step was to measure the embeddedness of a tie, defined as the number of common contacts of its two ends. To compare ties between athletes with a different number of contacts we standardize it with the overlap, which is the number of common contacts divided by the number of different contacts that the two athletes have together, not counting themselves. The overlap goes from 0 to 1, and it is high when the tie sits inside a dense group.

2.6. Permutation test

In this part of the analysis we use permutation tests to answer the research question. For each reciprocal tie we attach the level and the status of both ends. From the embeddedness of the ties we then compute, for each athlete, the mean overlap of their own ties. Athletes with a single tie are left out, because that tie has no common contact by construction. The tests compare high and low athletes, by commitment level and by status. For the local bridges we compare the share of bridging ties, shuffling the labels inside each club. As a robustness check we repeat the comparison excluding athletes with a single tie, because their only tie is a local bridge by construction, and shuffling the labels inside each club and degree. For the embeddedness we compare the mean overlap, and here the labels are shuffled only among athletes of the same club and with the same degree, so that a difference cannot be explained by the number of ties an athlete has. After each shuffle we compute the difference again, and the p-value is the share of shuffles that give a difference at least as large as the observed one, in absolute value.

Let $i = 1, \ldots, n$ be the athletes, y_i the outcome of athlete i, and $\ell_i \in \{\text{high}, \text{low}\}$ the label, either commitment level or status. The outcome is the share of bridging ties of the athlete for the local bridges, and the mean overlap of the athlete's own ties for the em-

beddedness. The observed statistic is the difference between the means of the two groups:

$$D_{\text{obs}} = \frac{1}{|H|}\sum_{i\in H} y_i \;-\; \frac{1}{|L|}\sum_{i\in L} y_i, \tag{1}$$

where $H = \{i : \ell_i = \text{high}\}$ and $L = \{i : \ell_i = \text{low}\}$.

The athletes are divided into strata $S_1, \ldots, S_K$. For the local bridges each stratum is a club, and for the embeddedness each stratum is a club with a given degree. For $b = 1, \ldots, B$ the labels are randomly permuted inside each stratum, which gives the permuted groups $H^{(b)}$ and $L^{(b)}$ and the statistic

$$D^{(b)} = \frac{1}{|H^{(b)}|}\sum_{i\in H^{(b)}} y_i \;-\; \frac{1}{|L^{(b)}|}\sum_{i\in L^{(b)}} y_i. \tag{2}$$

The two-sided p-value is

$$p = \frac{1 + \sum_{b=1}^{B} \mathbb{1}\left\{\, |D^{(b)}| \geq |D_{\text{obs}}| \,\right\}}{B+1}, \tag{3}$$

with $B = 5000$ permutations.

2.7. Temporal Cross-Validation and Data Leakage Prevention

In this section we want to predict whether the weekly running frequency of an athlete increases from month t to month $t+1$, and to see if knowing the contacts of the athlete at month t improves the prediction. The way the data are prepared matters as much as the models, because a feature that contains information from the future would give results that look good and mean nothing.

The target is built by moving the running frequency of the next month into the row of the athlete, separately for each athlete in each club. An athlete is eligible only if she or he is in the club in both months and is not already at the top of the scale, since the frequency cannot increase from there. The target is 1 if the frequency increases and 0 otherwise, and it is left empty for the athletes who are not eligible, so they are never used by a model. All the features are computed with information up to month t. For this reason the commitment level and the status, which use the whole year, are replaced by versions that are cumulated up to t. We also check this with a test: we rebuild the features as if the data ended at an earlier month, and the values for the months before must not change.

For the validation we cannot mix the months at random, because a model would see the future of the athletes it has to predict. Instead, we use an expanding window. The test set is a single month, from the sixth to the eleventh, and the model is trained only on the transitions whose outcome is already known at that month. The last month of the training set is used to choose the number of trees or epochs, so the test month is never used for this choice.

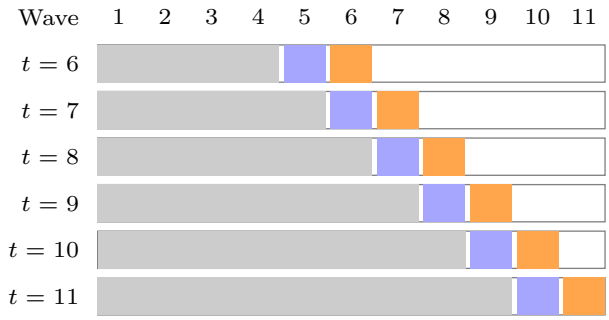


Figure 3: Expanding-window validation. Each row is a fold. Gray: training transitions; blue: last training transition, used to choose the number of trees or epochs; orange: test month; white: future months, never used.

2.8. Machine Learning models implementation

This section describes how the machine learning models were implemented. The aim is to test whether the increase in running behavior can be predicted with a machine learning approach. The first step was to define a benchmark to compare with the other models. For this we use a base rate model (Table 4), which takes as input only the outcomes of the training rows $\mathcal{T}$ and gives the same probability to every athlete.

The second benchmark is a logistic regression (second row of Table 4), chosen as a simple machine learning model. Its input is the vector of features $\mathbf{x}_i$ of athlete i at month t, standardized with training statistics and with no information on the contacts, and its output is the probability $\hat{p}_i \in [0, 1]$ that the running frequency increases.

The third model is XGBoost Chen and Guestrin (2016), a gradient boosting algorithm, which we use as a more advanced technique to predict the outcome. Its input $\mathbf{z}_i$ contains the features of athlete i at month t, summaries of the contacts, the club and the calendar month, and its output is $\hat{p}_i$. The number of trees K is chosen on the last training month.

The last model is a deep learning model, GraphSAGE Hamilton et al. (2017a), a graph neural network (Table 4). Its input is the snapshot graph of the club at month t with the features $\mathbf{x}_i$, where $\mathcal{N}_t(i)$ are the contacts of athlete i in that month. We use three versions of the edges: real, none and rewired. The output is $\hat{p}_i$ for every athlete of the graph.

Finally, we compare the models using the ROC-AUC and the PR-AUC on each of the six folds, to see which of them ranks the athletes best as the training window grows, and to answer the main question. Differences between models are assessed with a paired bootstrap on the athletes.

3. Results

3.1. Network Analysis: Strava as aggregation social structure

This section will cover the measurements from the conducted analysis, to answer a part of our research question: "to what extent does Strava work as an infrastructure for social aggregation?" Before discussing about the results, it is fundamental to consider that the clubs 1 and 4 are too small to conduct a scientific analysis, so in this results they will not be considered. In the first part, we conducted a network statistic by clubs, analyzing different methodologies and measures:" Ties, density, mean degree, Mean months per tie Reciprocity (annual), Isolated athletes, Largest component (share), Transitivity, Assortativity by commitment level. The results for the **Size and density** evidence a clubs range from 9 to 159 athletes this allow us to understand how is the high the variability and How they can influence the statistic.However **Mean kudos degree** (ties per athlete) still remains about 17 in club 2 and 14 in club 5, against club 3 to 4 .According with Figure2 the density falls with size 0.036 in club 3, the largest. This effect is consequences of the fact much bigger is the club and lower density is.

The results regarding the **Reciprocation** show in four of the five clubs about 79% to 83% of kudos ties belong to a pair that exchanged kudos in the same month, and annual reciprocity is 0.82 to 0.85. Although this trend is highly diffused the Club 3 differs with only about 50% of ties belong to a mutual pair (annual reciprocity 0.56), and ties last on average 3.9 months against 6.3 to 7.5 elsewhere

The results of the Core-periphery show how Club 3 has 15 athletes without any kudos tie and 74 of 159 (47%) without any strong tie; its largest strong-tie component contains only 52% of the athletes. In the other club the strong-tie component includes 93% (Club 2) of the athletes.

Strong ties are highly clustered: in the reciprocal graph the transitivity is between 0.54 and 0.72 in four clubs and 0.36 in Club 3 (Table 5). Paths are measured inside the largest component, which includes from 52% to 93% of the athletes, and they are short:

Table 4: Models used to predict whether the running frequency of athlete i increases from month t to month $t+1$ ($y_i = 1$). σ is the logistic function and $\hat{p}_i$ the predicted probability.

Model	Formula	Input	Output
Base rate	$\hat{p}_i = \frac{1}{\lvert\mathcal{T}\rvert} \sum_{k \in \mathcal{T}} y_k$	none; only the outcomes of the training rows $\mathcal{T}$	the same probability for every athlete
Logistic regression	$\hat{p}_i = \sigma(\beta_0 + \boldsymbol{\beta}^\top \mathbf{x}_i)$	features $\mathbf{x}_i$ of athlete i at month t, standardized with training statistics; no information on contacts	$\hat{p}_i \in [0,1]$
XGBoost	$\hat{p}_i = \sigma\left(\sum_{k=1}^{K} f_k(\mathbf{z}_i)\right)$, f_k regression trees	$\mathbf{z}_i$: features of athlete i at month t, summaries of the contacts, club and calendar month	$\hat{p}_i$; the number of trees K is chosen on the last training month
GraphSAGE (GNN)	$\mathbf{h}_i^{(0)} = \mathbf{x}_i, \quad l = 1,2$ $\mathbf{h}_i^{(l)} = \mathrm{ReLU}\big(\mathbf{W}_1^{(l)} \mathbf{h}_i^{(l-1)} + \mathbf{W}_2^{(l)} \,\mathrm{mean}_{j \in \mathcal{N}_t(i)} \mathbf{h}_j^{(l-1)}\big)$ $\hat{p}_i = \sigma\big(\mathbf{w}^\top \mathbf{h}_i^{(2)} + b\big)$	snapshot graph of the club at month t with features $\mathbf{x}_i$; $\mathcal{N}_t(i)$ are the contacts of i in that month; three versions of the edges: real, none, rewired	$\hat{p}_i$ for every athlete of the graph

Table 5: Structure of the kudos graph and of the reciprocal graph in each club. Months per tie: mean number of months with kudos. Assortativity: by commitment level. Mutual pairs: share of kudos ties inside a pair that exchanged kudos in the same month. Component: share of athletes in the largest component.

	Club 1	Club 2	Club 3	Club 4	Club 5
Athletes	27	58	159	9	76
Kudos graph					
Density	0.165	0.304	0.036	0.403	0.190
Months per tie	7.379	7.517	3.876	7.448	6.335
Reciprocity	0.845	0.816	0.556	0.828	0.826
Assortativity	0.011	0.004	0.048	−0.020	0.094
Reciprocal graph					
Mutual pairs	0.828	0.794	0.497	0.828	0.806
Isolated athletes	9	4	74	1	10
Component	0.593	0.931	0.522	0.889	0.816
Transitivity	0.606	0.542	0.355	0.720	0.535
Mean path length	1.708	1.835	3.254	2.036	1.988

Table 6: Permutation tests on high and low athletes. D is the mean of the high group minus the mean of the low group; the p-value is two-sided, with 5000 permutations. Labels are shuffled within club or within club and degree, as indicated.

Grouping	D	p
Bridging share, ≥ 1 tie (within club)		
Status	−0.232	< 0.001
Commitment level	−0.024	0.609
Bridging share, ≥ 2 ties (within club, degree)		
Status	+0.034	0.631
Commitment level	+0.046	0.126
Mean overlap, ≥ 2 ties (within club, degree)		
Status	+0.058	0.734
Commitment level	+0.039	0.875

the mean length is between 1.7 and 2.0 steps in every club except Club 3, where it is 3.3 and the diameter is 8. In the other four clubs, a behavior passing through strong ties would need few steps to cross the main component.

Assortativity by commitment level is close to zero in all the clubs, between −0.02 and 0.09, so athletes do not sort strongly by how much they run.

3.2. Local bridges, embeddedness and athlete status

For the local bridges, comparing all athletes with at least one tie, the difference between high- and low-status athletes is −0.23 ($p < 0.001$). This difference comes mostly from athletes with a single tie: 27 of the 29 are low-status, and their only tie is a local bridge by construction. After excluding them ($n = 202$) and shuffling the labels within club and degree, the difference is +0.034 ($p = 0.63$) for status and +0.046 ($p = 0.13$) for commitment level. For the embeddedness, the difference in mean overlap is +0.058 for status ($p = 0.73$) and +0.039 for commitment level ($p = 0.88$). With the number of ties held fixed, we find no difference between high- and low-status athletes, in bridging or in embeddedness. The low-status group is small (62 athletes with at least two ties, 7 of them in Club 3), so the power of these tests is limited.

3.3. Prediction models Assessment

According to the Table7 all models batted the base rate, therefore next-month increases in running frequency are partly predictable (top-decile lift of about 1.5 to 1.7), however the real contacts factor did add additional accuracy performance to the model, in fact the GNN with the real kudos graph did not show batter performance than the same GNN without any edges (ROC difference -0.018, 95% CI [-0.035, -0.001]; PR-AUC difference +0.006, CI [-0.015, +0.028]) or with randomly rewired edges (ROC -0.002, CI [-0.016, +0.011]). A simple model on the athlete's own history is as good as, or better than, any model using the network. The logistic regression shows significantly better performance than the real-edge GNN (ROC difference -0.036, CI [-0.055, -0.018]), and XGBoost, which receives explicit exposure variables, does not improve on it. However we need also to consider the fact that the uncertainty is large, each test month has about 300 observations (single-month ROC standard error about 0.04), so month-by-month differences should not be over-interpreted.

In this predictive set-up, the information carried by an athlete's contacts at month *t* does not add to what the athlete's own recent behavior already tells about next month's change. This is *consistent with*, but does not prove, the absence of social influence: the data are small and observational, the outcome is a coarse ordinal category, and influence from contacts may act over longer lags or through channels not recorded here.

Table 7: Out-of-time ROC-AUC by test month.

Month	Base	GNN-none	GNN-real	GNN-rew.	Logit	XGB
6	0.500	0.644	0.633	0.635	**0.700**	0.558
7	0.500	**0.674**	0.637	0.664	0.672	0.667
8	0.500	0.651	0.666	0.655	**0.681**	0.672
9	0.500	0.616	0.584	0.594	**0.652**	0.600
10	0.500	0.687	0.654	0.666	0.703	**0.734**
11	0.500	**0.682**	0.656	0.638	0.666	0.673
Mean	0.500	0.659	0.638	0.642	**0.679**	0.651

4. Limitations and Improvement

Several limitations should be acknowledged. First, the analysis is constrained by the number of clubs and their size, as two clubs have fewer than 30 athletes. Furthermore, each test month contains approximately 300 observations, which limits statistical power and may also affect neural network performance, given the need for larger training samples and more extensive individual-level data. Second, the operationalisation of key variables introduces further limitations: kudos are used as a proxy for social ties and recognition, while 'novice' captures tenure on Strava rather than actual running experience. The outcome is an ordinal weekly activity category ranging from 0 to 7, with a ceiling that may reduce the model's ability to distinguish among highly active athletes. Third, the observational nature of the data prevents us from distinguishing social influence from selection into ties, as similarities between connected athletes may arise because individuals with similar behaviours are more likely to form connections. Finally, the temporal structure may not fully capture the dynamics of social influence. Monthly observations and a one-month prediction horizon may overlook slower processes, while the annual aggregation adopted in Part II obscures the timing of network ties and changes in social relationships.

Acknowledgments

We would like to thank Professor Pierluigi Zama Ramirez for his guidance and valuable feedback throughout the development of this project. His suggestions contributed to the refinement of our analytical approach and the overall development of the research.

References

Ronald S. Burt. Structural holes and good ideas. *American Journal of Sociology*, 110(2):349–399, September 2004. doi: 10.1086/421787.

Damon Centola. The spread of behavior in an online social network experiment. *Science*, 329:1194–1197, 09 2010. doi: 10.1126/science.1185231.

Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, KDD '16, page 785–794, New York, NY, USA, 2016. Association for Computing Machinery. ISBN 9781450342322. doi: 10.1145/2939672.2939785. URL https://doi.org/10.1145/2939672.2939785.

Rob Franken, Hidde Bekhuis, and Jochem Tolsma. Kudos make you run! how runners influence each other on the online social network strava. *Social Networks*, 72, 2022a. doi: 10.1016/j.socnet.2022.10.001.

Rob Franken, Hidde Bekhuis, and Jochem Tolsma. Running together: How sports partners keep you running. *Frontiers in Sports and Active Living*, 4:643150, 2022b. ISSN 2624-9367. doi: 10.3389/fspor.2022.643150. URL https://www.frontiersin.org/journals/sports-and-active-living/articles/10.3389/fspor.2022.643150.

Mark S. Granovetter. The strength of weak ties. *American Journal of Sociology*, 78(6):1360–1380, 1973. doi: 10.1086/225469. URL https://uchicago.edu.

Will Hamilton, Zhitao Ying, and Jure Leskovec. Inductive representation learning on large graphs. In I. Guyon, U. Von Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc., 2017a. URL https://proceedings.neurips.cc/paper_files/paper/2017/file/5dd9db5e033da9c6fb5ba83c7a7ebea9-Paper.pdf.

William L. Hamilton, Rex Ying, and Jure Leskovec. Inductive representation learning on large graphs.

In *Advances in Neural Information Processing Systems 30 (NIPS 2017)*, pages 1024–1034, Long Beach, CA, USA, 2017b.

Miller McPherson, Lynn Smith-Lovin, and James M. Cook. Birds of a feather: Homophily in social networks. *Annual Review of Sociology*, 27(1):415–444, 2001. doi: 10.1146/annurev.soc.27.1.415.

Hayley C. Russell, Charlie Potts, and Emma Nelson. "if it's not on strava it didn't happen": Perceived psychosocial implications of strava use in collegiate club runners. *Recreational Sports Journal*, 47(1):15–25, 2023. doi: 10.1177/15588661221148170. URL https://doi.org/10.1177/15588661221148170.

Tom A.B. Snijders, Gerhard G. van de Bunt, and Christian E.G. Steglich. Introduction to stochastic actor-based models for network dynamics. *Social Networks*, 32(1):44–60, 2010. ISSN 0378-8733. doi: https://doi.org/10.1016/j.socnet.2009.02.004. URL https://www.sciencedirect.com/science/article/pii/S0378873309000069. Dynamics of Social Networks.

Li Zeng, Zack W. Almquist, and Emma S. Spiro. Stay connected and keep motivated: Modeling activity level of exercise in an online fitness community. In Gabriele Meiselwitz, editor, *Social Computing and Social Media. Technologies and Analytics*, pages 137–147, Cham, 2018. Springer International Publishing. ISBN 978-3-319-91485-5. URL https://rdcu.be/y5un4aMV7zJa.